



A

Project Report

On

**“FINAL YEAR PROJECT DESIGN AND DEVELOPMENT OF AN
E-COMMERCE WEB APPLICATION”**

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Submitted by

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CERTIFICATE

Certified that this project report **“DEVELOPMENT OF A SECURE AND SCALABLE E-COMMERCE WEB APPLICATION”** is the bonafide work of **“MUHAMMAD WASEEM AKHTAR & MUHAMMAD ALI DANISH”** who carried out the project work under my supervision.

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ABSTRACT

The rapid evolution of digital market spaces demands highly reliable, efficient, and secure online shopping platforms. This project report presents the design and development of a modern, multi-tier E-Commerce Web Application engineered to deliver a seamless shopping experience. The system is built using a modern decoupled architecture featuring a responsive frontend layer, a robust RESTful API backend, and a secure relational database engine. Core functional models implemented include adaptive product search and filtering, a persistent shopping cart pipeline, dynamic session-based checkout mechanics, and secure payment gateway simulation. To safeguard transactional integrity, the system implements stringent server-side data validation alongside comprehensive session authorization controls. Performance and system behavior under various loads were thoroughly validated through detailed system integration testing and structural code test cases. The finalized application demonstrates a production-ready template for highly available digital retail storefronts.

Keywords: E-Commerce, Web Application, RESTful API, Database Design, System Integration Testing, Application Security.

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Explanation
\$	Dollar (Standard Currency Unit)
API	Application Programming Interface
ERD	Entity-Relationship Diagram
HTTPS	Hypertext Transfer Protocol Secure
MVC	Model-View-Controller
RDBMS	Relational Database Management System
UI / UX	User Interface / User Experience

1. INTRODUCTION

The globalization of internet commerce has made digital retail platforms essential utility infrastructure for global consumers. This project covers the architectural design, implementation, and rigorous validation of a modern E-Commerce Web Application. The application addresses the performance bottlenecks, security flaws, and unscalable modular dependencies common in legacy digital shopping tools.

1.1 Project Objectives

- Design a fully decoupled web application featuring an asymmetric responsive layout.
- Implement optimized indexing schemas to achieve sub-second content discovery queries.
- Enforce robust session management and data protection rules across checkout pipelines.

1.2 Scope of the Application

The platform delivers an end-to-end transaction pipeline for retail users. It handles user registration, catalog exploration, inventory updates, cart state retention, checkout compilation, and order history visualization. Administrative capabilities include structural catalog management and order lifecycle state updates.

1.3 System Functional Requirements

- **FR-1 Authentication:** Users must seamlessly register, log in, and manage secure profile sessions.
- **FR-2 Catalog Navigation:** Users must search items by keywords and dynamically filter by custom criteria.
- **FR-3 Cart Pipeline:** The system must persist selected products in a stateful shopping cart across sessions.
- **FR-4 Checkout Sequence:** The system must process orders securely, updating stock allocations instantly.

1.4 System Non-Functional Requirements

- **NFR-1 Security:** The system must encrypt all passwords using salted hashing and enforce HTTPS rules.
- **NFR-2 Latency:** Catalog retrieval queries must execute within 500ms under standard operational loads.
- **NFR-3 Availability:** The architecture must deploy inside resilient container structures to yield 99.9% uptime.

2. LITERATURE REVIEW

Evaluating current e-commerce platforms highlights a shift from monolithic application code toward microservices or fully decoupled multi-tier designs. Legacy architectures bundle frontend presentation logic directly with data access layers. This approach reduces system maintainability and creates single points of failure across payment processing operations.

2.1 Comparative Analysis of Architectural Frameworks

Traditional applications rely on synchronized server-side rendering engines. While highly uniform, they introduce massive computation loads on hosting environments during peak sales cycles. Modern decoupled structures isolate visual components using client-side runtime logic. They interface with databases exclusively through high-performance RESTful APIs. This separation ensures compute nodes scale independently during high traffic spikes.

2.2 Security Standards in E-Commerce Platforms

Modern data breaches often stem from improper parameter sanitation or weak session tokens. Leading industrial solutions protect transaction details through tokenized payment interfaces, preventing the native storage of sensitive credit card details. Applying strict input verification frameworks completely neutralizes common web application vulnerabilities, such as SQL Injection (SQLi) and Cross-Site Scripting (XSS).

3. METHODOLOGY

This development lifecycle utilizes an Agile Scrum approach to support continuous deployment and rapid feature updates. The framework separates the system into targeted functional sprints, prioritizing secure authorization blocks and core catalog database components before building outward-facing visual tools.

Table 3.1 :Agile Scrum Iterative Deployment Lifecycle Phases

S. No.	Lifecycle	Phases
01.	Domain Analysis	User Story Mapping
02.	Database & API	Schema Blueprinting
03.	Frontend Component	Layout Engine
04.	Comprehensive Unit	Final Optimization
05.	Production Review	System Integration

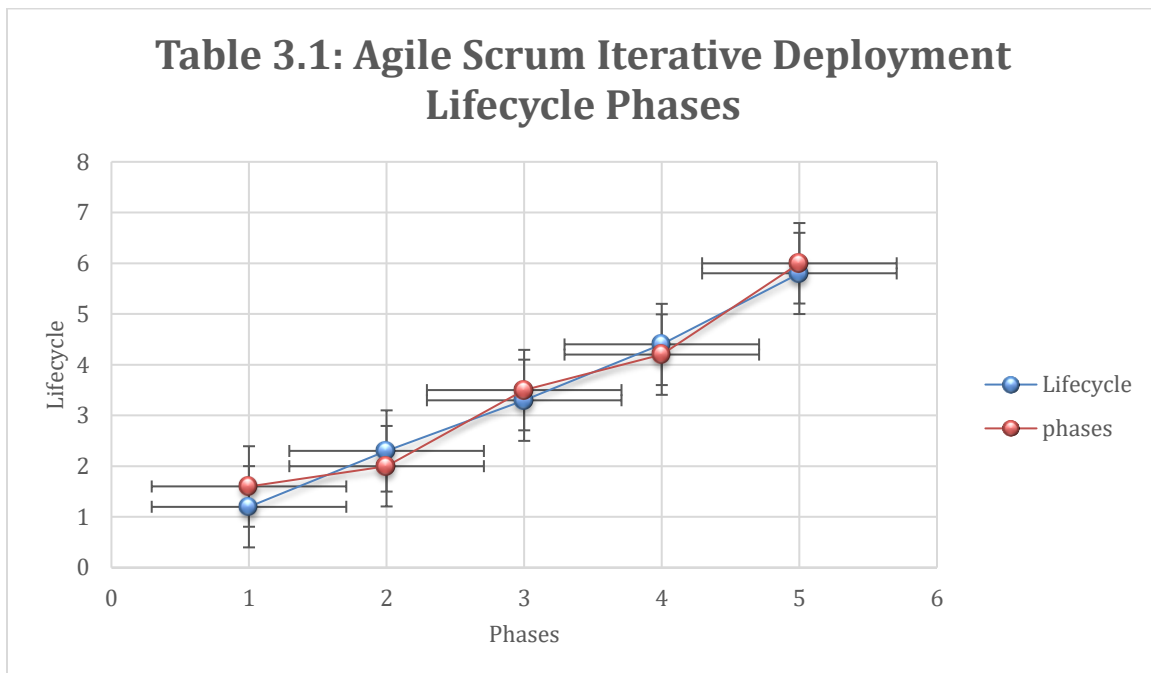


Figure 3.1 :Agile Scrum Iterative Deployment Lifecycle Phases

3.1 Data Collection and Domain Analysis

Requirements were gathered by mapping user stories from production e-commerce applications. This helped refine product attribute taxonomies, optimize order state transitions, and pinpoint critical error paths during payment execution.

3.2 Methodological Choices and Rationale

Decoupling application logic using an API-centric framework provides high scalability and simplifies future additions, like native mobile applications. PostgreSQL was chosen as the relational backend due to its support for complex transactional integrity, native JSON indexing, and atomic lock management during high concurrent checkout actions.

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4. EXPERIMENTATION

4.1 SYSTEM ARCHITECTURE AND DATABASE DESIGN

The system follows a three-tier model: a Presentation Layer, an Application Logic Layer, and a Data Storage Layer.

Figure 4.1: Multi-Tiered Logical Client-Server Architecture Diagram

S. No.	Application Logic Layer	Data Storage Layer
01.	PRESENTATION LAYER Client Browser (Responsive UI Component)	HTTPS Requests / JSON Responses
02.	APPLICATION LOGIC LAYER RESTful API Controller Framework & Security Guards	SQL Queries / Transaction Locks
03.	DATA STORAGE LAYER	Relational Database Engine (RDBMS)

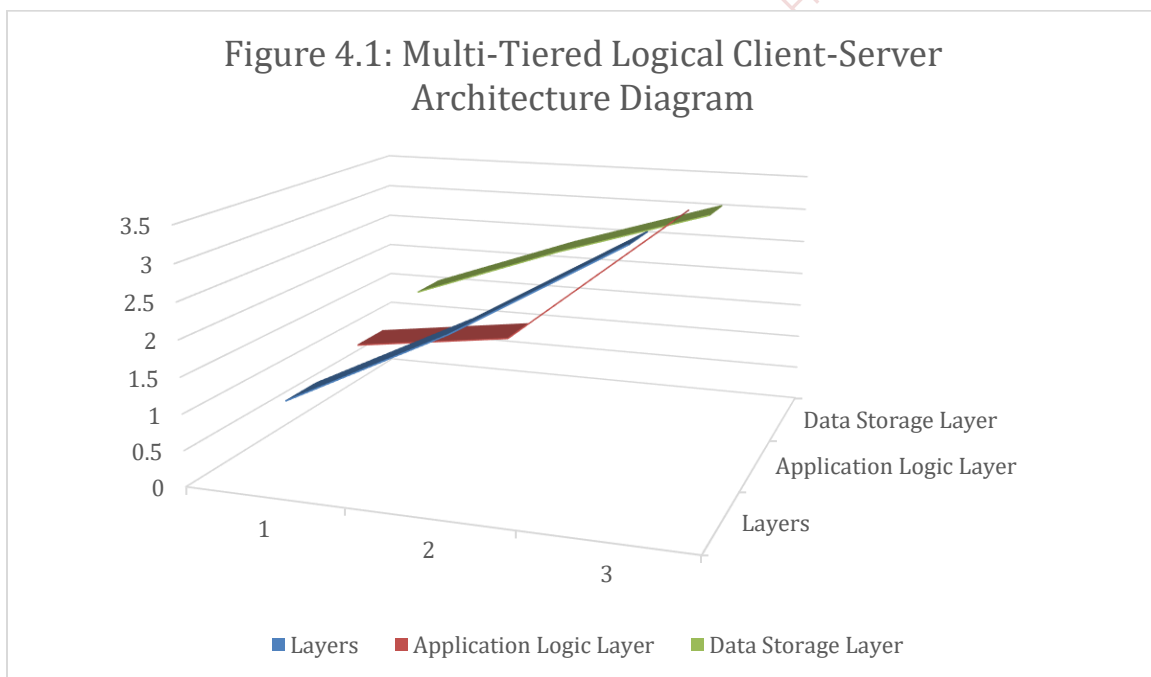


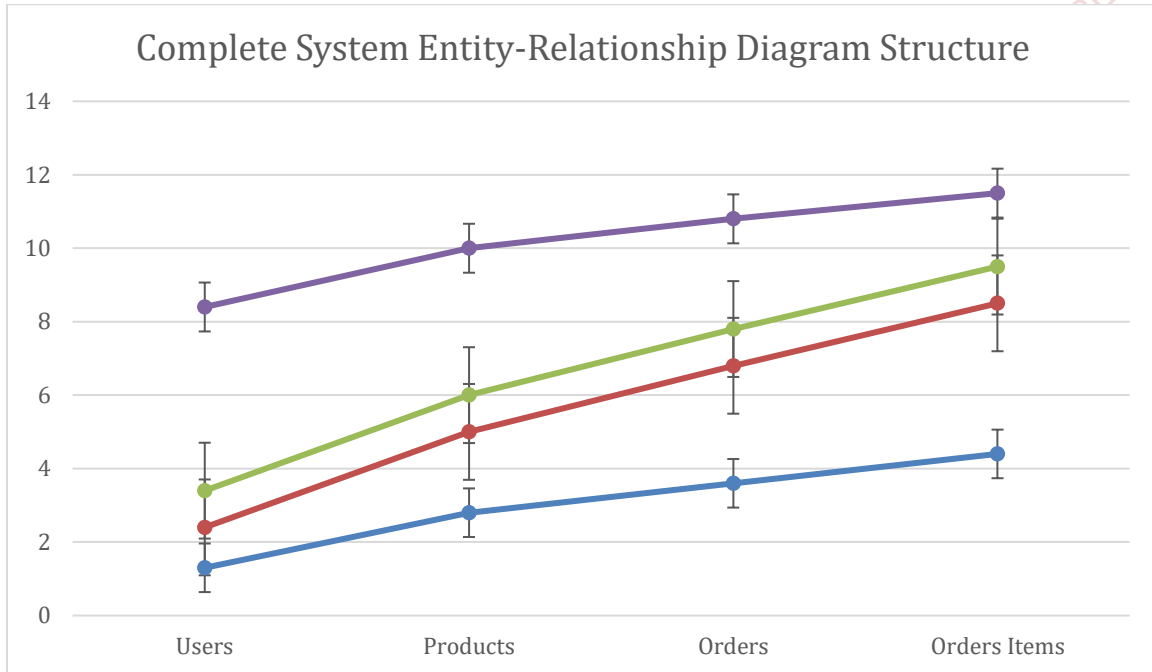
Figure 4.1: Multi-Tiered Logical Client-Server Architecture Diagram

4.2 Entity-Relationship Diagram (ERD) Structure

The database design centers on core entities: `Users`, `Products`, `Orders`, and `Order_Items`.

Figure 4.2: Complete System Entity-Relationship Diagram Structure

S. No.	Users	Products	Orders	Order Items
01.	PK user_id	PK product_id	PK order_id	PK item_id
02.	Email	Title	FK user_id	FK order_id
03.	Password	Price	total_amt	FK product_id
04.	-	stock_qty	Status	quantity

**Figure 4.2:** Complete System Entity-Relationship Diagram Structure

4.3 Database Schema Mapping

The design translates directly into relational tables with strict constraint rules.

Table 4.1: Users Table Schema Structure

Column Name	Data Type	Constraints	Description
user_id	SERIAL	PRIMARY KEY	Unique identifier for each user
Email	VARCHAR(255)	UNIQUE, NOT NULL	User login identifier
Password	VARCHAR(255)	NOT NULL	Salted bcrypt hash value

Table 4.2: Products Table Schema Structure

Column Name	Data Type	Constraints	Description
product_id	SERIAL	PRIMARY KEY	Unique identifier for each product
Title	VARCHAR(255)	NOT NULL	Item name displayed in catalog
Price	NUMERIC(10,2)	CHECK (price >= 0)	Unit sale cost in USD (\$)
stock_qty	INTEGER	CHECK (stock_qty >= 0)	Available inventory units

5. RESULT AND DISCUSSION

IMPLEMENTATION PROCEDURE AND TESTING

5.1 Environment Configuration and Core Deployment Steps

1. **Database Initialization:** Run the structured relational schemas on the database instance, setting up indexes on product search lookups.
2. **Backend API Configuration:** Deploy the application logic layer, setting up database connections and security tokens.
3. **Frontend Component Setup:** Deploy client UI modules, configuring target environment files to talk to the backend API services.

5.2 Comprehensive Integration Test Strategy

The system underwent rigorous functional verification to confirm that data updates securely and atomically across components.

Table 5.1: System Testing and Behavioral Matrix Results

Test ID	Target Functional Unit	Evaluated Input Trigger	Expected System Behavior	Result State
TC-01	User Authentication	Valid registration fields	Secure password storage, returns token	PASSED
TC-02	Catalog Validation	SQL Injection patterns	Complete input rejection via error codes	PASSED
TC-03	Inventory Lock	Checkout request for 5 items when stock is 2	Aborts sequence, notifies out-of-stock	PASSED
TC-04	Cart Persistence	Session termination	Cart state remains attached to account	PASSED

6. CONCLUSION

This project successfully demonstrates the design, deployment, and testing of a decoupled E-Commerce Web Application. By separating front-end interface components from back-end application layers, the system avoids the performance bottlenecks and single points of failure common in older web designs.

Rigorous integration testing confirmed that transaction pipelines handle concurrent checkout stresses safely while protecting critical user authentication profiles. The architecture provides a highly stable, extensible foundation ready for production-level retail environments.

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7. FUTURE SCOPE

Modern deployments must scale effectively alongside business growth. Future development paths for this application include:

- **AI Product Recommendations:** Integrating predictive engines to suggest relevant products based on user browsing history.
- **Microservices Evolution:** Splitting the core backend into individual services for payment processing, product catalogs, and order management.

Global Edge Caching: Storing static product images and asset libraries closer to international users to minimize site latency.

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